

BUSINESS REQUIREMENTS SPECIFICATION

Credit Risk Analytics — Bank Loan Portfolio Dashboard

Version 1.0 · Prepared by: Nishant Kulkarni · Date:25/08/ 2025

1. Document Purpose

This Business Requirements Specification (BRS) defines the business objectives, stakeholder requirements, scope, KPI definitions, and data rules for the Credit Risk Analytics Dashboard. It serves as the reference document for design, development, and validation of the Power BI reporting solution.

2. Business Problem Statement

The credit analysis team lacks a centralised, real-time view of loan portfolio health. Key decision-making is currently done using fragmented Excel reports which are manually updated, error-prone, and unable to surface risk patterns across borrower segments in a timely manner.

Specific pain areas identified through stakeholder interviews:

- No single view of funded vs. received loan amounts across the portfolio
- Cannot quickly identify the proportion of 'Bad Loans' (defaulted/charged-off) vs 'Good Loans'
- No visibility into Debt-to-Income (DTI) distribution by loan grade or employment length
- Month-on-month trend analysis requires manual effort — takes 2–3 days per reporting cycle
- Senior stakeholders cannot drill down by state, purpose, or borrower profile without IT support

3. Project Objectives

- Provide a compliance-ready, self-service Power BI dashboard for credit risk monitoring
- Enable senior stakeholders to make evidence-based lending decisions using live portfolio data
- Reduce manual reporting effort by automating KPI calculation and visualisation
- Surface anomalies and default indicators for early risk intervention

4. Stakeholders

Stakeholder	Role	Interest in Dashboard
Credit Risk Manager	Primary User	Portfolio-level risk visibility
Credit Analysts	Daily Users	Borrower-level drill-down, DTI filters
Senior Leadership	Consumers	Executive summary, trend overview
Compliance Team	Reviewers	Audit-ready data accuracy, loan status tracking

5. Scope

5.1 In Scope

- Total loan application volume, funded amount, and received amount KPIs
- Good Loan vs Bad Loan classification and percentage breakdown
- MTD (Month-to-Date) and MoM (Month-on-Month) trend tracking
- Borrower segmentation by: loan grade, employment length, loan purpose, home ownership, state
- DTI distribution analysis across loan grades
- Default prediction indicators and anomaly flagging

5.2 Out of Scope

- Real-time data streaming (dashboard refreshes on scheduled batch basis)

- Individual borrower PII details beyond anonymised segmentation
- Loan approval workflow automation
- Integration with external credit bureau APIs

6. KPI Definitions & Business Rules

KPI	Definition	Calculation	Business Rule
Total Applications	Count of all loan applications	COUNT(loan_id)	Include all statuses
Total Funded Amount	Sum of all disbursed loan amounts	SUM(loan_amount)	Funded ≠ Applied amount
Total Received Amount	Sum of repayments received	SUM(total_payment)	Payments to date only
Good Loan %	Loans with status: Fully Paid or Current	Good Loans / Total × 100	Current = not yet defaulted
Bad Loan %	Loans with status: Charged Off	Bad Loans / Total × 100	Charged Off = unrecoverable
Avg DTI	Mean Debt-to-Income ratio	AVG(dti) per segment	Higher DTI = higher risk

7. User Stories

The following User Stories define functional requirements from the perspective of each stakeholder:

US-001

As a: Credit Risk Manager

I want to: see a summary of total applications, funded amount, received amount, and average DTI on a single summary page

So that: I can assess overall portfolio health at a glance without navigating multiple reports

Acceptance Criteria: Summary KPIs visible on Dashboard page 1; values match source data within 0.1% variance

US-002

As a: Credit Analyst

I want to: filter loan data by grade (A–G) and employment length

So that: I can identify which borrower segments carry the highest default risk

Acceptance Criteria: Slicer filters apply across all visuals on the page simultaneously; Grade A–G and employment buckets available

US-003

As a: Senior Leadership

I want to: see month-on-month trends for funded amounts and default rates

So that: I can identify deteriorating portfolio trends before they escalate

Acceptance Criteria: MoM trend line chart with % change indicator; data visible for last 12 months minimum

US-004

As a: Compliance Officer

I want to: view loan status distribution (Fully Paid / Current / Charged Off) with drill-down by state

So that: I can prepare audit-ready summaries and flag high-risk geographic concentrations

Acceptance Criteria: Status breakdown table exportable to Excel; state-level map visual with colour-coded risk tiers

US-005

As a: Credit Analyst

I want to: view a detailed loan-level grid with filters for purpose, home ownership, and verification status

So that: I can investigate specific anomalies surfaced by the summary dashboard

Acceptance Criteria: Detail page with row-level data; minimum 10 filter dimensions available; supports export

8. Data Requirements

Field	Source	Notes
loan_id	Loan Management System	Primary key; unique per application
loan_amount / funded_amount	LMS	May differ; both required
loan_status	LMS	Values: Fully Paid, Current, Charged Off
dti	Credit Bureau Feed	Decimal; validated range 0–100
grade / sub_grade	Internal Scoring	A–G with sub-tiers
issue_date	LMS	Used for MoM trend calculation
addr_state	Application Form	2-letter US state code

9. Assumptions & Constraints

- Source data is available in CSV format and does not require real-time API connection
- Data refresh will occur on a scheduled basis (daily/weekly) — not live streaming
- Dashboard will be accessed via Power BI Desktop and Power BI Service
- All monetary values are in USD
- Borrower PII will not be included; data is anonymised at source before ingestion

10. Document Sign-off

This BRS is prepared as part of an independent analytics project by Nishant Kulkarni, demonstrating end-to-end BA documentation capability including requirement elicitation, stakeholder mapping, KPI definition, and User Story writing aligned to Agile SDLC practices.